

WASP-12b

R-0552

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-12_180117 · created 2026-07-22T10:21:52+00:00

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2458134.67272 +/- 0.00047 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.1219 +/- 0.0081
Transit depth (Rp/Rs)^2	1.49 +/- 0.2 [%]
Orbital Inclination (inc)	78.96 +/- 1.96
Airmass coefficient 1 (a1)	1.0227 +/- 0.0088
Airmass coefficient 2 (a2)	-0.0197 +/- 0.0042
Scatter in the residuals of the lightcurve fit is	0.86 %
Best Comparison Star	#3 - [356, 239]
Optimal Aperture	3.74
Optimal Annulus	6.9
Transit Duration (day)	0.1129 +/- 0.0072

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.1219 $\pm$ 0.0081 vs 0.1178 (deviation 0.49 σ)
Duration fit vs published	0.1129 d vs 0.125 d (deviation 1.64 σ)
Mid-time O-C	-1572.5 min
Depth SNR	14.7
Residual scatter	0.86 %

Light curve

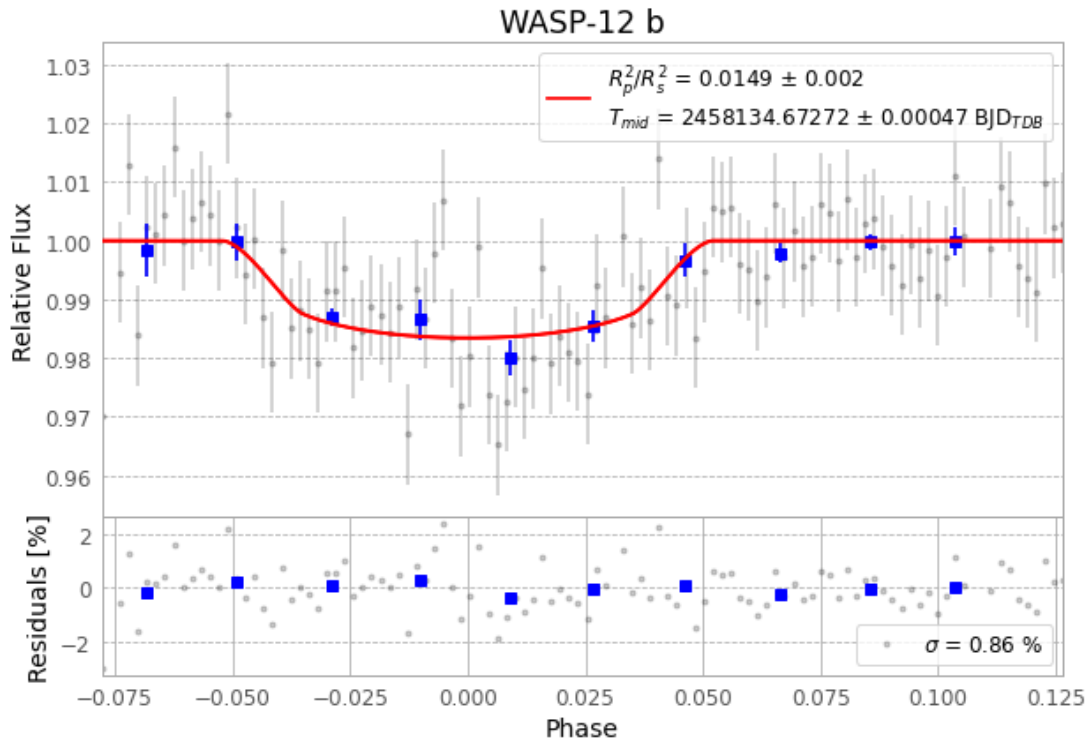


Figure 1 — FinalLightCurve_WASP-12 b_2018-01-15.png (SHA-256 9b6df39469157a95...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [373, 208], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 26406b6125e3ab0e...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit b365494

Data provenance

108 FITS frames (71 MB), WASP-12180116015712.FITS → WASP-12180116071812.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-12-180117.log (be10252b8cbc...), FinalLightCurve_WASP-12 b_2018-01-15.png (9b6df3946915...), FinalLightCurve_WASP-12 b_2018-01-15.csv (12a757112ed4...)

Compute resources

Wall time 135.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-22 10:21Z.